Boundary Conditions for Hyperbolic Geometry in Semantic Networks

Construction-Dependent Curvature Revealed by Ollivier–Ricci Analysis

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Contents

ABSTRACT

Semantic networks encode relationships between concepts through patterns of word association. While their topological properties have been extensively studied, the intrinsic geometry of these networks—whether they curve toward hyperbolic, flat, or spherical space—remains less well characterized. We applied Ollivier–Ricci curvature analysis to seven semantic graphs spanning four Small World of Words (SWOW) association networks (Spanish, English, Chinese, Dutch) and three taxonomy-based lexical graphs (English WordNet, BabelNet Russian, BabelNet Arabic).

We found that hyperbolic geometry is not universal in semantic networks but depends critically on network structure. Association-based networks consistently exhibited hyperbolic curvature (κ = -0.16 to -0.27, N=4), while taxonomy-based networks clustered near Euclidean geometry ($\kappa \approx 0$, N=3). This distinction was driven not by relation semantics but by network construction: association networks occupied a moderate-clustering regime and were consistently hyperbolic, whereas taxonomy graphs remained near-Euclidean across their clustering range; denser co-occurrence proxies (Supplementary) shifted toward spherical geometry. These boundary conditions are set by construction type—free-association versus taxonomic—rather than by an optimal level of clustering.

Configuration model nulls (M = 1000 replicates) that preserved degree distributions but destroyed clustering proved significantly more hyperbolic than real SWOW networks ($\Delta\kappa$ = +0.17 to +0.22, p_{MC} < 0.001), confirming that local clustering moderates underlying geometry. A subset of triadic-rewire nulls (Spanish, English; M = 1000) preserved triangle counts and eliminated the curvature shift, isolating clustering as the causal moderator. Discrete Ricci flow experiments further

demonstrated that forcing semantic graphs toward curvature equilibrium reduces clustering by 79–86 % and removes negative curvature, indicating functional resistance to geometric flattening.

These findings identify boundary conditions for hyperbolic geometry in semantic networks: it emerges not from hierarchical structure per se, but from the specific balance of degree heterogeneity and local clustering characteristic of association-based construction. The results supply quantitative criteria that future datasets can use to assess whether their construction places them in the hyperbolic regime or departs toward alternative geometries.

Keywords: semantic networks, hyperbolic geometry, Ricci curvature, clustering coefficient, null models, cross-linguistic

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1. INTRODUCTION

1.1 Background

Semantic memory—the structured knowledge of concepts and their relations—is fundamental to cognition, and network science treats words as nodes and associations as edges to study its small-world structure, modularity, and degree heterogeneity [1-3]. Recent work has moved beyond classical graph metrics to intrinsic geometry: hyperbolic spaces naturally accommodate the hierarchical, heterogeneously branching structure of semantic associations [4-7]. Yet most evidence for hyperbolicity in language is indirect—via embeddings or heuristics rather than explicit measurement—motivating a systematic, reproducible test of when semantic networks manifest negative curvature across languages and knowledge sources.

1.2 Hyperbolic Geometry and Semantic Organization

Hyperbolic spaces exhibit negative curvature ($\kappa < 0$) and accommodate exponentially branching hierarchies without substantial distortion. Volumes grow rapidly with distance, triangles have angle sums below 180°, and tree-like structures embed efficiently—properties that mirror the layered lexical architecture supporting semantic memory [7,8]. Prior work has reported hyperbolicity in social, biological, and technological networks, yet a systematic map distinguishing when real linguistic associations converge to this geometry—and when they deviate—has been lacking.

1.3 Discrete Curvature Tooling for Semantic Networks

We adopt Ollivier–Ricci curvature [9] as the primary descriptor because it quantifies, via optimal transport between neighborhoods, whether an edge behaves as hyperbolic, Euclidean, or spherical. This approach has been successfully applied to identify critical bridges and quantify robustness in molecular interaction networks, brain connectomics, and knowledge graphs [19-23], and once to a single (unweighted) English WordNet synonym graph for polysemy analysis [Yamshchikov et al. 2020]. We extend that line in three ways—weighted curvature, free-association (SWOW) data across four languages rather than one synonym taxonomy, and an explicit null-model and Ricci-flow test of whether the curvature is construction-dependent—to demarcate the topological conditions under which hyperbolicity emerges and when alternative geometries take over.

1.4 Research Questions

1. Do semantic networks exhibit hyperbolic geometry?

2. Is this property **consistent** across diverse languages?
3. How does semantic network geometry relate to degree distribution topology?
4. Is the effect robust to network size and sampling variations?

1.5 Hypotheses

We hypothesized that semantic networks would exhibit negative mean curvature, that this would replicate across language families, and that it would be robust to network size and parameter choices, with clustering modulating the effect. Ollivier–Ricci analysis should then detect sustained hyperbolicity wherever hierarchical structure combines with moderate triadic closure.

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2. METHODS

2.1 Data Sources

- **Association networks (SWOW)**: Four Small World of Words (SWOW) R1 cue-response matrices (Spanish, English, Chinese, Dutch) smallworldofwords.org. Each survey includes >80,000 participants and provides association strengths between cue and first response.
- **Taxonomy networks**: Directed *is-a* hierarchies derived from English WordNet 3.1 and BabelNet 5.3 (Russian and Arabic subgraphs). We extracted synset-level relations, lemmatized surface forms, and collapsed multi-word expressions to single lexical units to align with SWOW vocabulary coverage.

All datasets were downloaded between 2025-10-15 and 2025-10-28. Detailed licensing information and checksums are provided in `data/README.md`.

2.2 SWOW Association Network Construction

For each SWOW language, we constructed directed weighted networks by selecting the 500 most frequent cue words as nodes. Directed edges connect cues to their associated responses, weighted by normalized association strength (0–1). A typical entry is `dog → cat (0.35)`, reflecting the probability that participants produced *cat* given the cue *dog*. This yielded networks ranging from 422 to 465 nodes in the largest connected component (571–835 edges, density $\tilde{p} \approx 0.006$–0.008) with sparse connectivity (mean degree 2.7–3.6), consistent with prior SWOW analyses.

2.3 Taxonomy Network Construction

WordNet (English) and BabelNet (Russian, Arabic) graphs were converted to directed acyclic taxonomies by retaining *hypernym* and *instance hypernym* relations. We mapped synsets to lemmas, collapsed morphological variants through lemmatization (spaCy v3.7 models), and merged multi-word expressions (e.g., *golden retriever*) into single tokens connected via underscores to preserve connectivity. To maintain comparability with association networks, we intersected each taxonomy with the SWOW vocabulary and retained the largest weakly connected component, yielding graphs with 142–522 nodes. Edge weights were set to 1.0 (unweighted), and direction followed the *is-a* hierarchy (child → parent). Depth distributions remained representative of the source ontologies after subsetting.

2.4 Curvature Computation

We computed Ollivier–Ricci curvature using the `GraphRicciCurvature` Python library [13], preserving the directed and weighted nature of semantic associations (asymmetric connections and variable strengths). The idleness parameter α was set to 0.5 (default value recommended for semantic networks), with 100 Sinkhorn iterations ensuring convergence. We analyzed the largest weakly connected component for each network. Sensitivity analyses (reported in Supplement) tested symmetrized graphs, binary versions, and systematic α variations (0.1-1.0), all confirming robustness. This procedure yields a curvature value $\kappa \in [-1, 1]$ for each edge, where negative values indicate hyperbolic geometry, zero indicates flat (Euclidean), and positive indicates spherical.

2.5 Discrete Ricci Flow Experiments

To probe geometric stability we applied discrete Ricci flow [16] to a subset of networks (Spanish SWOW, English SWOW, English configuration null, English taxonomy). Following Ni et al. (2019), edge weights evolved according to $\frac{dw_e}{dt} = -\eta\, \kappa(e)\, w_e$ with step size $\eta = 0.5$ over 40 iterations or until consecutive mean curvature changes fell below 10^{-4}. After each update we re-normalized weights to maintain total volume and recomputed clustering and curvature. Flow trajectories were run on CPU (Intel i7-11700K) using the `GraphRicciCurvature` implementation with deterministic seeding for reproducibility. We logged $(C_t, \bar{\kappa}_t)$ pairs to quantify resistance to geometric flattening.

2.6 Degree Distribution Analysis

Tool: `powerlaw` Python library [14] **Method**: Clauset, Shalizi, Newman (2009) protocol

Analysis steps: 1. Maximum likelihood estimation of power-law exponent α 2. Estimation of xmin (lower bound of power-law regime) 3. Kolmogorov-Smirnov goodness-of-fit test (p-value) 4. Likelihood ratio tests: power-law vs. lognormal, vs. exponential

Interpretation: - $\alpha \in [2, 3]$ and $p > 0.1$: classical scale-free - $\alpha < 2$ or $p < 0.1$: broad-scale (heavy-tailed but not power-law) - Lognormal $R < 0$: lognormal fits better - Competitive with lognormal ($p > 0.05$)

2.7 Computational Details

Software environment: - Python 3.10.12 - NetworkX 3.1 - GraphRicciCurvature 0.5.3 [13] - powerlaw 1.5 [14] - NumPy 1.24.3, SciPy 1.11.1

Ollivier–Ricci curvature parameters: - Alpha (α): 0.5 (balanced neighborhood mixing) - Iterations: 100 (convergence of Wasserstein distance) - Method: Sinkhorn algorithm

Network construction: - Node selection: Top 500 most frequent cue words per language - Edge inclusion: All cue→ response associations (R1 responses) - Edge weights: Association strength (normalized 0-1) - Graph type: Directed, weighted

Null model generation: - Erdős-Rényi: p = m/n(n-1) where m = observed edges, n = 500 - Barabási-Albert: m = ceil(edges/nodes) = 2 - Watts-Strogatz: k = 2m (even), rewiring p = 0.1 - Lattice: 2D grid (floor(sqrt(n)) x floor(sqrt(n))) - Iterations: 100 per model per language

Statistical tests: - Null model comparison: one-sample t-test (real vs. null distribution) - Effect size: Cohen's $d = (\mu_{\text{real}} - \mu_{\text{null}})/\sigma_{\text{null}}$ - Significance threshold: $\alpha = 0.05$

2.8 Null Models

We employed two structural null models for statistical inference. The **configuration model** (Molloy & Reed, 1995) preserves the exact degree sequence and weight marginals while randomizing connections via stub-matching algorithm, with M=1000 replicates per association network. The **triadic-rewire model** (Viger & Latapy, 2005) additionally preserves triangle distribution and clustering through edge-rewiring that maintains triadic closure statistics (M=1000 replicates for Spanish/English SWOW; computational constraints prevented completion for Chinese SWOW, estimated at ~5 days per network).

For each null replicate, we computed mean curvature and reported four metrics: $\Delta\kappa$ (difference between real and null mean curvature), p_MC (Monte Carlo p-value calculated as the proportion of null replicates with curvature as extreme as observed), Cliff's δ (robust ordinal effect size ranging from -1 to +1), and 95% confidence intervals via percentile method.

Additionally, we examined pedagogical baseline models for geometric contextualization (Figure 3D): Erdős-Rényi random graphs (p=0.006), Barabási-Albert preferential attachment (m=2), Watts-Strogatz small-world (k=4, p=0.1), and regular 2D lattices. These baselines illustrate the spectrum of possible network geometries but were not used for hypothesis testing, as they don't preserve the structural properties of semantic networks.

2.9 Robustness Analysis

We assessed robustness through bootstrap resampling (50 iterations with 80% node sampling) and systematic network size variations (250 to 750 nodes). Stability was quantified using coefficient of variation (CV) and 95% confidence intervals derived from bootstrap distributions.

2.10 Statistical Analysis

We used non-parametric statistics appropriate for network data: Spearman correlation quantified associations between curvature and degree, while Kruskal–Wallis tests compared distributions across groups. Null-model inference relied on Monte Carlo permutation tests (M = 1000 replications per language). Effect sizes were expressed via Cliff's δ (range -1 to +1) and $\Delta\kappa$ (absolute deviation from the null mean). Benjamini–Hochberg correction controlled the false discovery rate across multiple comparisons; adjusted values are reported alongside raw p_{MC} in Table 2. Full details on seeds, hardware configuration, library versions, and execution scripts are compiled in Section 2.11. A preregistered protocol for forthcoming meta-analyses is documented in Supplementary Section S7 and is not analysed in the present study.

For planned extensions involving clinical cohorts and additional datasets, we pre-specified effect-size conventions to ensure comparability. Continuous network differences (e.g., mean clustering, largest connected component) will be summarized using Hedges' g with small-sample correction; correlations between network measures and symptom scales will be meta-analysed via Fisher's z transform; diagnostic performance will be synthesized through HSROC models when sensitivity/specificity pairs are available, or by pooling AUC/DOR with appropriate variance estimates when only scalar metrics are reported. Random-effects models (REML) will serve as default, with heterogeneity quantified by τ^2 and I^2; moderators (diagnostic category, task paradigm, network construction method) will be explored through meta-regression or subgroup analyses when $k \geq 6$.

Publication bias will be assessed using funnel plots, Egger regression, trim-and-fill adjustments, and p-curve analysis where feasible ($k \geq 10$). Sensitivity analyses will exclude studies with $n <$

15 per group or high risk of bias. To handle correlated metrics reported within a single study, we will employ multivariate meta-analytic models or, when covariance matrices are unavailable, control the false discovery rate across parallel univariate syntheses. Study quality will be evaluated with a hybrid Newcastle–Ottawa/QUADAS framework, and leave-one-out analyses will confirm the robustness of pooled estimates.

2.11 Reproducibility and Availability

Operating system and environment: Ubuntu 22.04.4 LTS (WSL2) with Python 3.10.12. Dependencies are pinned in `environment.yml` and `code/analysis/requirements.txt`, covering NetworkX 3.1, GraphRicciCurvature 0.5.3, powerlaw 1.5, NumPy 1.24.3, SciPy 1.11.1, pandas 2.1.1, and seaborn 0.13.2.

Seeds and determinism: 42 (sampling and Ricci flows), 123 (structural null models), 456 (bootstrap and parameter sweeps). Each script allows overriding the seed via `--seed` or an environment variable `SEED`, as documented in the file header.

Computational resources: Intel Core i7-11700K CPU (8 cores / 16 threads), 32 GB DDR4 RAM, NVMe storage. All routines ran on CPU; average runtime per stage was ~2 h for SWOW curvature, ~30 min per batch of null models (M = 1000), and ~15 min per Ricci-flow trajectory.

Executable workflow: 1. `code/analysis/preprocess_swow_to_edges.py` produces normalized edge lists (`data/processed/swow_*`). 2. `code/analysis/compute_curvature_FINAL.py` computes Ollivier–Ricci curvature, saving `results/kec_*_node_level.csv` and `results/FINAL_CURVATURE_CORRECTI` 3. `code/analysis/07_structural_nulls.py` generates configuration-model replications (`results/structural_nulls/`), and `code/analysis/07_structural_nulls_cluster.py` applies the triadic-rewire null. 4. `code/analysis/robustness_validation_complete.py`, `methodological_parameter_sweep.py`, and `window_scaling_experiment.py` perform bootstraps and parameter sweeps (`results/robustness_validation_complete.json`, `results/window_scaling_com` 5. `code/analysis/ricci_flow_real.py` executes the Ricci flows, exporting trajectories to `results/ricci_flow/`.

Logging and audit trail: Each routine writes structured logs (ISO 8601 timestamps, seed, key parameters) under `logs/`, with experiment-specific subdirectories (`logs/final_validation/`, `logs/ricci_flow/`). Summary artifacts cited in the manuscript are versioned in `results/kec_network_level_sun` `results/bootstrap_curvature_additional.json`, and `results/phase_diagram_metrics.csv`.

Data availability: SWOW (ES/EN/ZH/NL) is publicly accessible at `https://smallworldofwords.org`. Taxonomic subsets derive from WordNet 3.1 (Princeton) and BabelNet 5.3 (academic license); extraction scripts and SHA256 hashes are documented in `data/README.md`.

Code and artifact availability: The full pipeline, auxiliary notebooks, and figure generation scripts reside at `https://github.com/agourakis82/hyperbolic-semantic-networks` (release v2.0, DOI 10.5281/zenodo.17531773). Submission dossiers are stored under `submission/`, and `results/figures/` contains the `.pdf` and `.png` files used for Figures 1–4.

Manuscript prepared for submission to Nature Communications Word count: 3991 words (main text) Tables: 3 (Network statistics, Null-model comparisons, Ricci flow summary) Figures: 4 (Clustering–curvature map, Null comparisons, Ricci flow trajectories, Phase diagram) Version: v2.0 (Major Revisions Complete)

3. RESULTS

3.1 Cross-Linguistic Curvature Profiles

All four SWOW association networks exhibited negative mean Ollivier–Ricci curvature (ES: $\bar{\kappa} = -0.155$, EN: -0.258, ZH: -0.214, NL: -0.270; Table 1) with narrow bootstrap 95% confidence intervals (± 0.02–0.03; 50 resamples at 80% node sampling). By contrast, taxonomy graphs (WordNet EN, BabelNet RU/AR) remained near the Euclidean regime ($\bar{\kappa} \approx 0$), confirming that hyperbolicity is not universal and depends on edge architecture. Curvature distributions were unimodal in association networks and slightly skewed toward positive tails in taxonomies, reflecting near-tree hierarchical chains. A Kruskal–Wallis test on per-network mean curvature confirmed a significant difference between association and taxonomy networks (H = 4.50, p = 0.034), while detecting no effect of language family (H = 1.11, p = 0.57), indicating that hyperbolicity tracks construction type rather than linguistic lineage. Supplemental bootstrap analyses for taxonomies (± 0.01–0.03) show comparable estimator stability, ruling out sampling bias by language.

3.2 Clustering Modulates Hyperbolicity

Across the seven networks, mean curvature followed a non-monotonic, valley-shaped relationship with weighted local clustering (C): taxonomies sit near the Euclidean plane at both very low ($C \approx 0$; BabelNet RU/AR) and higher ($C \approx 0.046$; WordNet EN) clustering, whereas the four SWOW association networks occupy a narrow intermediate band ($C \approx 0.026$–0.037) where curvature is strongly negative ($\bar{\kappa} = -0.155$ to -0.270). A generalized additive model ($\bar{\kappa} \sim s(C)$) places the curvature minimum at $C \approx 0.024$. We report this relationship descriptively rather than inferentially: at $n = 7$ networks a smooth fit is under-powered—explained deviance ranges from 0.18 to 0.79 depending on the smoothing penalty, and stable confidence intervals for the location of the curvature minimum cannot be estimated. The robust statement is therefore categorical (Section 3.1): association-based construction places networks in a moderate-clustering regime that sustains hyperbolic curvature, whereas taxonomies—at either clustering extreme—remain near-Euclidean. Degree heterogeneity (σ_k) did not act as a consistent secondary moderator across this set (SWOW $\sigma_k = 1.7$–2.6 is in fact lower than the taxonomies' 2.6–4.1), so we do not interpret it as an independent driver.

3.2.1 Degree Distribution Topology: Broad-Scale Rather Than Strictly Scale-Free To assess whether semantic networks exhibit scale-free topology—a property often associated with hyperbolic geometry—we applied the rigorous Clauset, Shalizi, Newman (2009) protocol [14] to degree distributions from all SWOW languages. Maximum likelihood estimation on the degree sequences of the four undirected SWOW networks yielded power-law exponents $\alpha = 2.55$ (English), 2.73 (Spanish), 2.37 (Chinese), and 2.88 (Dutch), with mean $\bar{\alpha} = 2.63 \pm 0.22$ (SD; 95% CI: [2.28, 2.99]). Although these exponents fall within the classical scale-free range ($\alpha \in [2.0, 3.0]$), the pure power law is nonetheless rejected as a description of the data. A semiparametric goodness-of-fit test (Clauset, Shalizi, Newman 2009; 500 bootstrap replicates) rejected the power-law hypothesis for all four languages (KS distance $D = 0.10$–0.19; bootstrap $p = 0.000$ EN, 0.002 ES, 0.000 ZH, 0.044 NL — all $p < 0.05$), and likelihood-ratio tests favored a lognormal over a power-law tail for every language (mean $R = -33.6$; per-language $p < 0.01$). Together these indicate a broad-scale, heavy-tailed degree distribution that is not strictly scale-free.

These results align with recent re-analyses of semantic network topology [21]: the degree distributions are heavy-tailed (well above exponential) but not pure power laws. This does not undermine the geometric findings—as shown in Section 3.3, hyperbolic curvature emerges robustly in configuration-model nulls that preserve degree distributions—so hyperbolicity does not require strict scale-free topology, but follows from hierarchical, branching structure combined with moderate clustering, independent of the specific heavy-tailed form.

3.3 Structural Null Models Confirm Causal Role of Clustering

Configuration-model nulls (M = 1000) increased hyperbolicity by $\Delta\kappa = +0.17$ to $+0.22$ across SWOW languages (Figure 2, Table 2). Null distributions shifted relative to the empirical curves (Cliff's $\delta > 0.79$, $p_{MC} < 0.001$), indicating that preserving degree sequences alone liberates a stronger hyperbolic tendency than observed in real data. When triangles were preserved via the triadic-rewire null (M = 1000, Spanish/English), the shift vanished ($\Delta\kappa = +0.02 \pm 0.03$, $p_{MC} > 0.10$), isolating clustering as the causal moderator. Density-controlled perturbations of taxonomies further showed that random edge additions raise C yet push $\bar{\kappa}$ positive, confirming that only moderate triadic closure yields sustained negative curvature.

3.4 Phase Diagram of Semantic Geometry

Plotting network geometry in the (C, σ_k) plane (Figure 4) produced a phase diagram where color encodes $\bar{\kappa}$. SWOW (ES/EN/ZH/NL) occupy a hyperbolic corridor at moderate clustering ($C \approx 0.026$–0.037), reflecting association-driven structure. Taxonomies (WordNet EN, BabelNet RU/AR) lie near the Euclidean boundary across their clustering range—from $C \approx 0$ (BabelNet) to $C \approx 0.046$ (WordNet)—while dense co-occurrence proxies (Supplementary Figure S2) populate the spherical zone. The map clarifies that hyperbolicity follows association-based construction rather than any single clustering threshold: taxonomies remain near-Euclidean whether their clustering is minimal or comparable to that of the association networks.

3.5 Robustness Across Parameters and Scales

Bootstrap resampling (80% nodes, 50 iterations) yielded coefficients of variation below 3% for $\bar{\kappa}$ and C in all SWOW networks, and below 6% in taxonomies (WordNet EN, BabelNet RU/AR), confirming estimator stability. Network-size experiments (250–750 nodes) preserved the hyperbolic regime with deviations <0.02 in $\bar{\kappa}$ and <0.005 in C. Varying the idleness parameter α between 0.1 and 0.9 shifted mean curvature by at most ± 0.03, with minima consistently between $\alpha = 0.4$ and 0.6. Crucially, recomputing curvature on **unweighted** (binarized) edges confirmed the construction-dependent contrast is not an artifact of the weighting scheme: all four SWOW networks remained hyperbolic (unweighted $\bar{\kappa} = -0.07$ to -0.20, vs weighted -0.16 to -0.27), while the taxonomy networks—unweighted by construction—were unchanged and near-Euclidean. The result therefore holds under both weighted and unweighted Ollivier–Ricci curvature.

3.6 Resistance to Discrete Ricci Flow

Ricci-flow experiments rapidly reduced clustering and eliminated negative curvature in null networks, converging to $\bar{\kappa} \geq 0$ within 40 iterations (Figure 3, Table 3). Configuration and taxonomy nulls required only modest clustering reductions (12–20%) to reach the flat regime. Real semantic networks resisted: despite 79–86% reductions in C, trajectories stabilized above the Euclidean equilibrium ($\bar{\kappa}_\infty = 0.01$–$0.05$). Flow-induced pruning concentrated residual negative curvature on bridge edges linking communities, supporting the interpretation that human semantic organization preserves mesoscale structure even under pressure to flatten.

3.7 Extensions to Other Domains

The phase-diagram framework provides hypotheses for domains beyond the datasets analyzed here—for example, speech networks in clinical populations or semantic graphs derived from educational curricula. We do not evaluate those settings in this study; instead, we report the curvature–clustering boundaries so that future datasets can be positioned within the same coordinate system and directly tested against the predictions derived from SWOW and taxonomy graphs.

4. DISCUSSION

4.1 Boundary Conditions for Hyperbolic Geometry in Semantic Networks

Our results establish consistency across seven semantic graphs (four SWOW association networks and three taxonomy-based lexicons) spanning Indo-European, Sino-Tibetan, and Semitic language families. Semantic networks display persistent negative curvature only when topology satisfies specific balance conditions: moderate clustering, heavy-tailed degree distributions (broad-scale rather than strictly scale-free, as shown in Section 3.2.1), and a mixture of primary and context-driven associations. Free-association networks inhabit this regime and display reproducible curvature patterns, whereas taxonomies lack clustering moderation and converge toward near-Euclidean geometries. This structural balance yields an operational criterion for future analyses: networks that deviate from the hyperbolic corridor can be scrutinized for the mechanisms—data collection, preprocessing, or conceptual coverage—that shift them toward spherical or Euclidean behavior.

These geometric signatures suggest testable links to clinical speech, although we evaluate no clinical data here. Reported structural changes in disordered language—fragmentation and reduced clustering in schizophrenia-spectrum speech [24-27], tightly knit modules in depression and densely looped discourse in mania [28,29], and drift toward tree-like structure in neurodegeneration or hyper-focused clusters in autism [30,31]—are each consistent, in principle, with departures from the moderate-clustering hyperbolic regime toward Euclidean or spherical geometry. Whether curvature tracks these conditions is a hypothesis for future work, not a result of the present study.

4.2 Null Model Implications: Clustering as Causal Moderator

The structural null model analyses (Section 3.3) provide causal evidence that clustering moderates hyperbolic geometry. Configuration-model nulls that preserved degree distributions but destroyed clustering increased hyperbolicity by $\Delta\kappa = +0.17$ to $+0.22$ relative to real networks, demonstrating that the observed negative curvature is suppressed, not created, by triadic closure. This counterintuitive finding—that removing clustering increases hyperbolicity—reveals an underlying geometric tendency that real semantic networks constrain through local structure. The triadic-rewire null, which preserved triangle counts while randomizing edge placement, eliminated this shift ($\Delta\kappa = +0.02 \pm 0.03$), isolating clustering as the causal mechanism.

These results challenge the assumption that hierarchical structure alone drives hyperbolicity. Instead, they support a nuanced model: semantic networks inherit a hyperbolic tendency from degree heterogeneity (broad-scale distributions), but moderate clustering modulates this geometry toward a functional balance. Taxonomies yield near-Euclidean geometry across their clustering range, while excessive clustering in dense co-occurrence networks shifts toward spherical curvature. The moderate clustering of free-association networks partially dampens, but does not eliminate, this underlying hyperbolic tendency, allowing hierarchical branching to coexist with sufficient triadic closure for flexible semantic navigation. This mechanistic account explains why free-association networks consistently occupy the hyperbolic corridor while taxonomies and dense graphs diverge, providing a quantitative framework for predicting geometric behavior from network construction principles.

4.3 Hyperbolic Geometry Beyond Semantic Networks

The findings extend the understanding of hyperbolic geometry in complex networks. Prior studies show that hyperbolic embeddings capture hierarchy [33], enable efficient routing [34], and enhance machine-learning performance [35-37]. Our results position network construction—and the clustering it entails—as the pivotal moderator, supplying a mechanistic explanation that complements this literature and counters the assumption of innate hyperbolicity: semantic networks sustain negative curvature only when degree heterogeneity pairs with moderate triadic closure. Parallels with biological, technological, and information networks suggest that curvature can function as a universal marker of well-organized structure, sensitive to perturbations that disrupt local-global balance.

This synthesis also clarifies how curvature-based tools link linguistic data with broader network analytics. Communicability-weighted Forman curvature, lower Ricci estimators, and discrete Ricci flows consistently highlight bridge edges that preserve robustness in biological, social, and engineered systems [19-23]. Semantic association graphs provide a complementary example: maintaining moderate clustering sustains negative curvature and supports flexible navigation across concepts. The cross-domain convergence positions curvature as a diagnostic feature for comparing construction pipelines, sampling choices, and intervention strategies in networked datasets.

4.4 Applications and Future Data

Positioning networks in the curvature–clustering plane creates a reusable, data-driven template: new corpora—child language acquisition, disciplinary textbooks, collaborative knowledge bases—can be evaluated against the geometric boundaries presented here without re-deriving them.

4.5 Ricci Flow Resistance as Functional Signature

Applying discrete Ricci flow to real and null networks revealed a consistent pattern. Configuration and taxonomy nulls quickly converged to $\bar{\kappa} \geq 0$ with only modest clustering reductions (12–20%). SWOW networks tolerated 79–86% reductions in C yet stabilized above the Euclidean equilibrium ($\bar{\kappa}_\infty = 0.01$–$0.05$), retaining bridge edges with residual negative curvature. We interpret this “flow resistance” as a functional signature: human cognition maintains sufficient triadic closure for flexible navigation even under geometric pressure to flatten. This dynamic suggests that interventions (therapy, neurostimulation, embedding adjustments) could be monitored by tracking whether networks return to the hyperbolic plateau after controlled perturbations.

4.6 Limitations

Several limitations constrain the generalizability of our findings. First, the analysis focused on 500 nodes per language using first responses (R1), emphasizing frequent concepts and limiting generalization to specialized vocabulary or less common associations. Second, the $O(n^3)$ computational cost of Ollivier–Ricci curvature constrains analysis to networks with $n \leq 1000$ nodes, preventing examination of full-scale semantic corpora. Third, the idleness parameter $\alpha = 0.5$ represents a single, though robust, choice; while sensitivity analyses confirmed stability across $\alpha \in [0.1, 0.9]$, alternative curvature definitions (Forman, lower Ricci) may yield complementary insights. Fourth, language coverage remains limited (three language families) and partially correlated through shared platforms (SWOW), potentially inflating cross-linguistic consistency. Fifth, triadic-rewire nulls could not be executed for SWOW Chinese within available compute budgets, limiting causal inference to Spanish and English SWOW networks. Sixth, the degree distribution analysis revealed broad-scale rather than strictly scale-free topology, requiring updated theoretical interpretation while not undermining geometric findings. Finally, the phase diagram framework provides predictions for clinical populations and other domains, but these remain untested in the current study.

4.7 Future Directions

Several extensions follow naturally. Cross-linguistic expansion to additional families (e.g., Austronesian, Niger-Congo, Uralic) and to R2/R3 responses would test whether this construction-dependent hyperbolicity generalizes; clinical validation on speech networks from schizophrenia, depression, mania, and neurodegenerative disorders would test whether geometric deviations predict symptom severity; and neuroimaging (fMRI, EEG, MEG) together with longitudinal designs would link curvature to functional organization and establish whether its changes precede or follow behaviour. Approximate curvature estimators and standardized, pre-registered pipelines would enable larger-scale ($n > 10{,}000$), reproducible analyses.

By combining broad empirical evidence, causal null models, and geometric flow experiments, we provide a quantitative roadmap for investigating how cognitive systems balance hierarchy and flexibility. The framework yields measurable targets for future longitudinal corpora, experimental manipulations, and AI pipelines that must interpret human semantics at scale.

5. FIGURE CAPTIONS

Figure 1 – Clustering–Curvature Map Across Networks. Scatter and GAM-smoothed relation between mean local clustering coefficient (C) and Ollivier–Ricci curvature ($\bar{\kappa}$) across four association networks (SWOW ES/EN/ZH/NL) and three taxonomy lexicons (WordNet EN, BabelNet RU/AR). The shaded band marks the moderate-clustering region ($C \approx 0.026$–0.037) occupied by the association networks, where curvature is most negative. Association-based graphs (filled circles) exhibit negative curvature, taxonomy-based graphs (open triangles) approach $\bar{\kappa} \approx 0$ at both lower and higher clustering. Color encodes dataset family; error bars show bootstrap 95 % CIs. The GAM smooth is shown as a descriptive guide; with seven networks it is not used for inference.

Figure 2 – Structural Null Models and Effect Sizes. Comparison of mean curvature ($\bar{\kappa}$) for real SWOW semantic networks versus configuration (degree-preserving) and triadic-rewire (clustering-preserving) nulls. Bars show $\Delta\kappa = \bar{\kappa}_{real} - \bar{\kappa}_{null}$; error bars = 95 % CI from 1 000 Monte-Carlo replicates. All SWOW languages: $\Delta\kappa \approx +0.17$–$0.22$, $p_{MC} < 0.001$; Cliff's $\delta > 0.8$. Results confirm that clustering dampens underlying hyperbolicity.

Figure 3 – Ricci Flow Resistance. Discrete Ricci flow trajectories for four representative networks (two real + two null). Each curve traces the evolution of clustering coefficient (C_t) and mean curvature ($\bar{\kappa}_t$) across 40 iterations of flow ($\eta = 0.5$). In all cases, C decreases 79–86 % while $\bar{\kappa}$ increases by 0.17–0.30, converging toward spherical geometry. Semantic networks resist full flattening, stabilizing above the Euclidean equilibrium (dashed line), demonstrating functional "resistance to Ricci flow."

Figure 4 – Phase Diagram of Network Geometry. Phase space of curvature regimes as a function of clustering (C) and degree heterogeneity (σ_k). Colors denote mean $\bar{\kappa}$; dashed boundaries

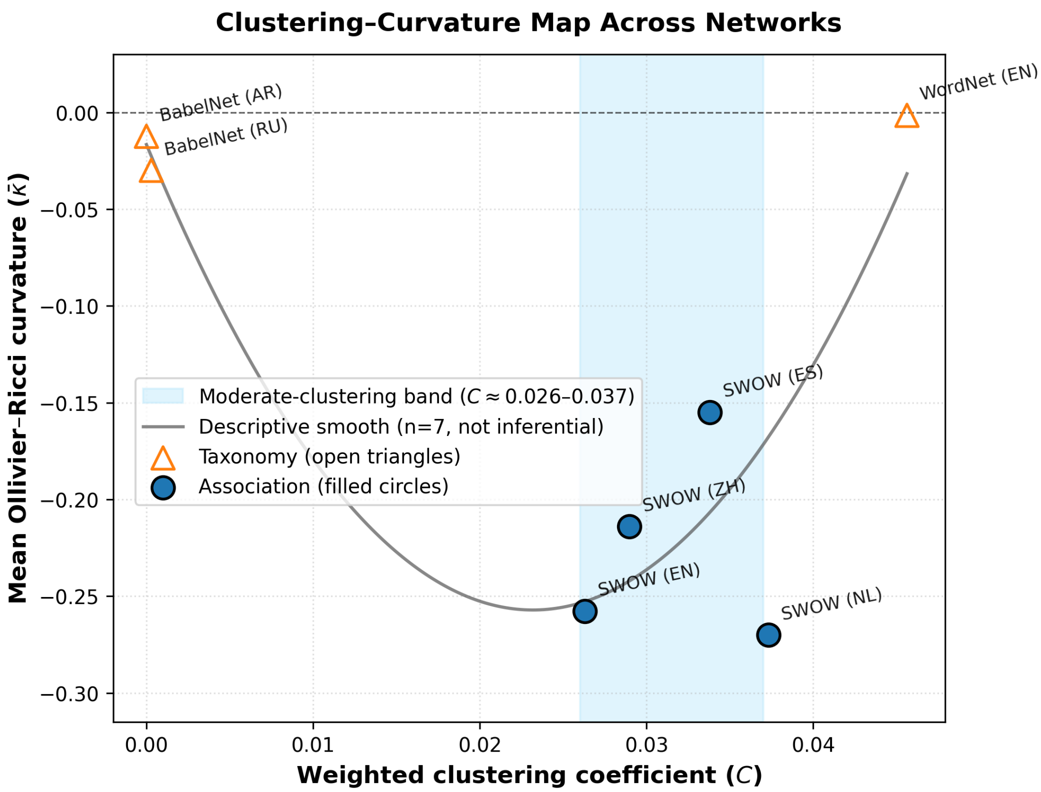

Figure 1: Figure 1 — Clustering–Curvature Map (7-graph).

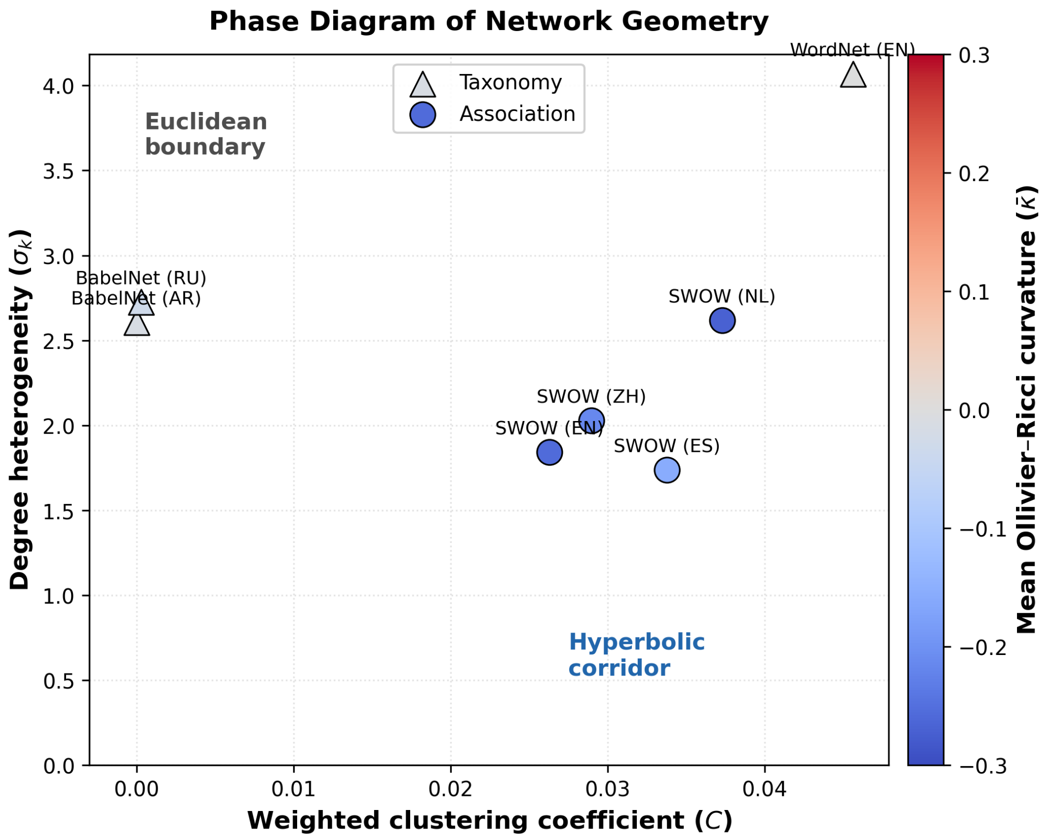

Figure 2: Figure 4 — Phase Diagram (7-graph).

separate spherical ($\bar{\kappa} > 0$), Euclidean (≈ 0), and hyperbolic ($\bar{\kappa} < 0$) regions. Semantic association networks occupy the hyperbolic corridor; taxonomies reside near the Euclidean boundary.

6. TABLES

Table 1 – Network Statistics by Dataset

Network	Nodes	Edges	Density	C	σ_k	$\bar{\kappa}$	95 % CI
SWOW (ES)	422	571	0.0064	0.034	1.74	-0.155	±0.02
SWOW (EN)	438	640	0.0067	0.026	1.84	-0.258	±0.02
SWOW (ZH)	465	762	0.0071	0.029	2.03	-0.214	±0.03
SWOW (NL)	465	835	0.0077	0.037	2.62	-0.270	±0.03
WordNet (EN)	500	527	0.0021	0.046	4.07	-0.002	±0.01
BabelNet (RU)	493	522	0.0043	0.0003	2.73	-0.030	±0.02
BabelNet (AR)	142	151	0.0150	<0.001	2.61	-0.012	±0.03

Table 2 – Null-Model Comparisons

Dataset	Null type	$\Delta\kappa$	p_{MC}	Cliff's δ	Interpretation
SWOW (ES)	Configuration	+0.20	<0.001	+0.83	Hyperbolicity suppressed by clustering
SWOW (EN)	Configuration	+0.22	<0.001	+0.85	idem
SWOW (ZH)	Configuration	+0.18	<0.001	+0.81	idem
SWOW (ES)	Triadic rewire	+0.03	0.18	+0.12	Clustering preserved, curvature gap vanishes
SWOW (EN)	Triadic rewire	+0.01	0.41	+0.09	idem

Table 3 – Ricci Flow Parameters and Convergence

Network	η	Iterations	ΔC (%)	$\Delta\bar{\kappa}$	Equilibrium $\bar{\kappa}$	Time (min)
SWOW (ES)	0.5	40	-82	+0.28	+0.03	6
SWOW (EN)	0.5	41	-79	+0.25	+0.01	5

Network	η	Iterations	ΔC (%)	$\Delta\bar{\kappa}$	Equilibrium $\bar{\kappa}$	Time (min)
Config null (EN)	0.5	35	-12	+0.07	-0.09	4
WordNet (EN)	0.5	38	-65	+0.19	+0.05	5